

INFINITY FITNESS STUDIOS

Data Validation & Dashboard Testing

Member Retention, Utilization & Operational Analytics
December 2025 - February 2026

Document Control	Details
Document ID	IFS-DVT-001
Version	1.0
Role	Data Analyst
Purpose	Define the validation approach for prepared analytical data and the functional testing required for the Power BI dashboard.
Scope	Member, membership/plan, attendance/check-in, payment, trainer, class/session, revenue and expense analysis; KPI calculations; filters; cohort views; dashboard reconciliation.
Evidence Status	Portfolio reconstruction of the validation/testing approach. No formal historical UAT execution log, defect counts or signed test evidence is claimed.
Confidentiality	No real member records, client values, screenshots containing restricted data or unsupported test outcomes are included.

1. Testing Objectives

- Confirm that source data is sufficiently clean and consistent for the intended analysis.
- Prevent duplicate records or incorrect joins from inflating member, attendance, payment, booking or session metrics.
- Validate date and time logic used for cohort, renewal, trend and utilization analysis.
- Confirm KPI calculations use the intended business definitions.
- Verify Power BI filters and visual interactions behave consistently.
- Reconcile dashboard outputs to the prepared analytical data.
- Protect confidential client/member information in portfolio-facing outputs.

2. Validation Layers

Layer	Focus	Examples
Source Validation	Input completeness and usability	Required fields, dates, numeric values, plan labels, identifiers
Transformation Validation	Cleaning and derived logic	Date standardization, joining month, time bands, flags, segmentation
Relationship Validation	Data-model joins and grain	Member-to-attendance, plan-to-membership, trainer-to-session relationships
KPI Validation	Business calculations	Retention, attrition, renewals, check-ins, revenue/expenses, trainer measures
Dashboard Validation	Report behavior	Filters, visuals, context, reconciliation, readability
Confidentiality Validation	Portfolio safety	No restricted member/client data or unsupported metrics exposed

3. Source Data Validation Checks

ID	Validation Area	Check	Expected Control
DV-01	Member identifier	Check for blanks, inconsistent IDs and duplicates that could affect distinct-member counts.	Member counts are not knowingly duplicated by obvious identifier issues.
DV-02	Joining dates	Validate usable date format and identify invalid/missing dates before cohort assignment.	Only valid joining dates feed joining-month cohorts.
DV-03	Renewal dates/evidence	Check date/status consistency before renewal calculations.	Renewal measures use valid, interpretable renewal evidence.
DV-04	Attendance records	Check member linkage, date/time usability and duplicate check-ins.	Valid attendance records feed utilization and engagement analysis.
DV-05	Plan labels	Standardize equivalent plan labels before comparison.	Monthly, 3-month, 6-month, annual and personal-training categories are consistently classified where present.
DV-06	Payment/revenue/expense amounts	Validate numeric type and obvious invalid/missing amounts before aggregation.	Only validated numeric values feed commercial totals.
DV-07	Trainer/session linkage	Check trainer IDs and session/booking relationships before trainer analysis.	Trainer measures are not knowingly inflated by broken joins.
DV-08	Missing activity evidence	Review blanks rather than automatically converting them to churn.	Missing evidence is not automatically treated as member attrition.

4. Transformation & Segmentation Validation

ID	Transformation	Validation
TV-01	Joining_Month	Derived month must correspond to the valid Joining_Date.
TV-02	Membership_Duration	Duration must use the documented start/end/activity context and must not produce unexplained negative durations.
TV-03	Retention_Flag	Flag must follow the agreed retained-member definition for the selected period.
TV-04	Attrition_Flag	Flag must use stopped-attendance and/or cancellation evidence according to the agreed rule; exact inactivity threshold must not be invented.
TV-05	Renewal_Flag	Flag must distinguish eligibility for renewal from actual renewal evidence.
TV-06	Time_Band / Hour	Derived value must correspond to the original valid check-in time.
TV-07	Cohort_Period_Index	Relative period must correctly represent time since joining cohort.
TV-08	Plan Segment	Equivalent plan names must map to the intended standardized plan category.

5. Relationship / Model Validation

- Compare row counts before and after joins to detect unexpected record multiplication.
- Confirm one member can link to multiple attendance/payment/session records without duplicating the member dimension.
- Confirm plan filtering reaches membership-related measures through the intended relationship.
- Confirm trainer filtering affects only linked trainer/session activity.
- Confirm date filtering produces consistent period context across applicable fact tables.
- Avoid ambiguous many-to-many paths unless the sanitized rebuild genuinely requires them.

6. KPI Calculation Validation

KPI	Validation Method	Key Risk
Retention Rate	Recalculate retained members ÷ eligible members for a selected sample/context.	Wrong denominator or inconsistent retained-member rule
Attrition Rate	Recalculate attrited members ÷ eligible members under the agreed attrition definition.	Treating missing data as churn or using an invented inactivity window
Renewal Rate	Compare renewed memberships with memberships eligible for renewal.	Counting non-eligible memberships in denominator
Active Members	Validate distinct-member logic under selected period/filter.	Duplicate member records
Cohort Retention	Check joining-month assignment, cohort base and subsequent-period retained/engaged counts.	Members assigned to wrong cohort or changing denominator
Total Check-ins	Compare dashboard count to prepared valid attendance records.	Duplicate check-ins
Revenue / Expenses	Reconcile Power BI aggregation to validated analytical totals.	Invalid amounts or incorrect classifications
Trainer Measures	Compare sessions/bookings/attendance to prepared trainer-linked records.	Broken linkage or double-counting

7. Dashboard Functional Test Cases

ID	Test	Action	Expected Result	Evidence Status
DT-01	Dashboard loads	Open the Power BI report using the prepared model.	Report opens and intended analytical visuals are available.	Not historically recorded
DT-02	Date filtering	Select a valid date/period filter.	Applicable KPIs and visuals respond to the selected period consistently.	Not historically recorded
DT-03	Plan filtering	Select a membership plan.	Relevant membership, renewal and plan visuals reflect the selected plan context.	Not historically recorded
DT-04	Cohort analysis	Select/review a joining cohort and subsequent periods.	Cohort membership and later-period engagement remain consistent with the prepared cohort logic.	Not historically recorded
DT-05	Attendance trend	Review attendance by date/hour/day.	Displayed counts reconcile to valid attendance records in the same context.	Not historically recorded
DT-06	Peak / underutilized periods	Compare time-based attendance views.	Higher/lower-use periods reflect the underlying check-in pattern; evening importance is visible where supported.	Not historically recorded
DT-07	Renewal metrics	Filter relevant period/plan and review renewal measures.	Renewal values reconcile to eligible and renewed membership records.	Not historically recorded
DT-08	Trainer filtering	Select a trainer where data supports it.	Sessions/bookings/attendance respond only to the selected trainer context.	Not historically recorded
DT-09	Cross-filter behavior	Interact with a chart/segment.	Related visuals respond consistently without unexpected metric inflation.	Not historically recorded
DT-10	Commercial totals	Review revenue/expense output for a selected period.	Values reconcile to validated prepared totals.	Not historically recorded
DT-11	Reset filters	Clear applied filters.	Dashboard returns to the intended default/all-data context.	Not historically recorded
DT-12	Confidentiality check	Review portfolio-facing report/output.	No restricted member-level data or unsupported confidential values are exposed.	Portfolio control

8. Known Finding Validation

The post-New-Year engagement decline should be validated through the same cohort and attendance logic used to identify it: joining-month cohorts, subsequent attendance/retention behavior, membership-duration segmentation and member activity over time. The public portfolio should describe the pattern qualitatively unless a sanitized dataset is available to reproduce a defensible numerical result.

9. Dashboard Reconciliation Procedure

Step	Procedure
1	Choose a controlled filter context, such as a selected period and plan.
2	Capture the relevant dashboard KPI or visual total.
3	Calculate the same result from the prepared analytical data using the same business definition.
4	Compare values and investigate differences caused by filters, joins, blanks, duplicates or business-rule interpretation.
5	Correct the data/model/measure if required and rerun the comparison.
6	Document unresolved limitations instead of forcing the numbers to match.

10. Defect Classification for Portfolio Testing

Severity	Definition	Example
Critical	Produces materially incorrect core business insight or exposes restricted data.	Retention calculation fundamentally wrong; confidential member data visible
High	Major KPI/filter/model behavior is incorrect.	Plan filter produces duplicated member counts
Medium	A non-core visual or calculation has a limited issue.	Label, sorting or secondary calculation issue
Low	Cosmetic/usability issue that does not change business meaning.	Spacing, title or formatting inconsistency

No historical defect counts, severities or closure statistics are claimed because a formal execution/defect log from the original engagement is not retained.

11. Portfolio Dashboard Image Control

The dashboard images created for the public portfolio are presentation mockups and must not be treated as screenshots of the client's historical production dashboard or as evidence of actual KPI values. Any illustrative numbers, dates, plan labels, member segments, facility zones or attrition reasons appearing in a mockup must be labeled as illustrative/sanitized before publication if they are not supported by the original client data.

12. Exit Criteria

- Core KPI calculations reconcile to the prepared analytical data for tested contexts.
- Filters and cross-filtering do not create unexplained metric changes.
- Cohort logic correctly assigns members to joining periods.
- Attendance/utilization views use valid date/time records.
- Known limitations are documented.
- Public portfolio outputs contain no restricted client/member data.
- No unsupported test pass rate, defect count or business-impact percentage is presented.

13. Requirements Traceability

Validation / Test Area	Related Requirements / Stories
Retention & Attrition	BR-02, BR-03, BR-11; US-01
Cohort Analysis	BR-04, BR-11; US-02
Attendance / Utilization	BR-05, BR-06; US-03
Renewal / Plan Performance	BR-07, BR-08, BR-12; US-04
Trainer Utilization	BR-09; US-05
Power BI Dashboard	BR-01, BR-13; US-06
Data Quality / Confidentiality	BR-14, BR-15; US-07

14. Testing Evidence Status

Evidence Type	Status
Formal historical UAT plan	Not established from retained evidence
Historical test execution results	Not retained
Historical defect log	Not retained
Signed client acceptance	Not established
Positive gym-manager feedback	Confirmed
Dashboard built from scratch	Confirmed
Portfolio validation specification	Documented in this artifact
Sanitized portfolio mockups	Created separately; illustrative values must not be represented as client facts